Multiplication and division of algebraic fractions

No common denominator needed — but factorise before you multiply, or you will multiply out work you are about to cancel.

LESSON 17–19 | 3 × 40 MIN | ALGEBRA 8, §5 | STAGE 9 · 8.3–8.4

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Multiply algebraic fractions and simplify the result.
- Divide by multiplying by the reciprocal.
- Cancel across the multiplication sign before multiplying out.
- Raise an algebraic fraction to a power.

TEXTBOOK

National	§5, pp. 27–29
Cambridge	Learner's Book pp. 175–184
Workbook	Workbook 8.3–8.4

01 Explanation

The two rules

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{A \cdot C}{B \cdot D} \qquad \frac{A}{B} : \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C} = \frac{A \cdot D}{B \cdot C}$$

Multiply straight across. To divide, **turn the second fraction upside down and multiply**. Nothing else changes.

For division you need $C \neq 0$ as well as $B \neq 0$ and $D \neq 0$ — you cannot divide by a fraction whose numerator is zero.

Cancel first, multiply last

THE HABIT THAT MAKES THIS TOPIC EASY

Factorise every numerator and denominator, cancel **anything on top against anything on the bottom**, and only then write down what is left. If you multiply out first you create a degree-4 polynomial that you must then factorise again.

$$\frac{x^2 - 9}{x} \cdot \frac{x^2}{x + 3} = \frac{(x - 3)(x + 3)}{x} \cdot \frac{x \cdot x}{x + 3} = x(x - 3)$$

The $(x + 3)$ cancels diagonally, and one x cancels with the other denominator. Two cancellations, no expansion.

Powers of a fraction

$$\left(\frac{A}{B}\right)^n = \frac{A^n}{B^n}$$

$\left(\frac{2x}{3y}\right)^3 = \frac{8x^3}{27y^3}$ — the index reaches the numbers as well as the letters.

Dividing by a whole expression

A whole expression is a fraction with denominator 1, so dividing by $x + 1$ means multiplying by $\frac{1}{x + 1}$:

$$\frac{x^2 - 1}{x} : (x + 1) = \frac{(x - 1)(x + 1)}{x} \cdot \frac{1}{x + 1} = \frac{x - 1}{x}$$

02 Worked examples

EXAMPLE 1 ***Simplify*** $\mathbf{\dfrac{2a}{a+b} : \dfrac{4a^2}{a^2-b^2}}$

1. $\dfrac{2a}{a+b} \cdot \dfrac{a^2-b^2}{4a^2}$

   Turn the second fraction over.

2. $a^2 - b^2 = (a - b)(a + b)$

   Factorise before multiplying.

3. $\dfrac{2a \cdot (a-b)(a+b)}{(a+b) \cdot 4a^2}$

   Now every factor is visible.

4. $\dfrac{a-b}{2a}$

   Cancel (a + b), and cancel 2a against $4a^2$.

ANSWER

$\dfrac{a-b}{2a}$, $a \neq 0, a \neq \pm b$

EXAMPLE 2 ***Simplify*** $\dfrac{x^2 - 5x + 6}{x^2 - 4} \cdot \dfrac{x^2 + 2x}{x - 3}$

1. $x^2 - 5x + 6 = (x - 2)(x - 3)$

   First numerator.

2. $x^2 - 4 = (x - 2)(x + 2)$

   First denominator.

3. $x^2 + 2x = x(x + 2)$

   Second numerator — common factor.

4. $\dfrac{(x - 2)(x - 3)}{(x - 2)(x + 2)} \cdot \dfrac{x(x + 2)}{x - 3}$

   Everything factorised.

5. *cancel* $(x - 2)$, $(x + 2)$ *and* $(x - 3)$

   Three cancellations, all diagonal.

ANSWER

$x, \;\; x \neq \pm 2, x \neq 3$

EXAMPLE 3 ***Simplify*** $\left(1 - \dfrac{1}{x}\right) : \dfrac{x-1}{x^2}$

1. $1 - \dfrac{1}{x} = \dfrac{x-1}{x}$

   Combine the bracket first.

2. $\dfrac{x-1}{x} \cdot \dfrac{x^2}{x-1}$

   Turn the divisor over.

3. *cancel* $(x - 1)$*, and* x *against* x^2

   Two cancellations.

4. x

   A surprisingly simple answer — typical of this topic.

ANSWER

$x,\ \ x \neq 0, x \neq 1$

03 Interactive model

Show this on the board and let a learner drive it.

Multiplying and dividing, step by step

$$\frac{x^2 - 9}{x} \cdot \frac{x^2}{x + 3}$$

1. $x^2 - 9 = (x - 3)(x + 3)$

Factorise before anything else.

2. $$\frac{(x - 3)(x + 3)}{x} \cdot \frac{x \cdot x}{x + 3}$$

Split x^2 so the cancelling is visible.

3. *cancel* $(x + 3)$ *and one* x

Both cancellations go diagonally.

4. $x(x - 3) = x^2 - 3x$

Multiply what is left.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Multiply	Ko‘paytirish	Умножение
Divide	Bo‘lish	Деление
Product	Ko‘paytma	Произведение
Quotient	Bo‘linma	Частное
Reciprocal fraction	Teskari kasr	Обратная дробь
Power of a fraction	Kasrning darajasi	Степень дроби
Cancel across	Krest bo‘ylab qisqartirish	Сократить накрест
Divisor	Bo‘luvchi	Делитель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{a}{b} : \frac{c}{d}$ equals:

- $\frac{ac}{bd}$
- $\frac{ad}{bc}$
- $\frac{bc}{ad}$
- $\frac{a + d}{b + c}$

$\frac{x}{2} : \frac{x}{4}$ equals:

- $\frac{1}{2}$
- 2
- $\frac{x^2}{8}$
- 8

$\left(\frac{3x}{2y}\right)^2$ equals:

- $\frac{3x^2}{2y^2}$
- $\frac{9x^2}{4y^2}$
- $\frac{6x^2}{4y^2}$
- $\frac{9x}{4y}$

Before multiplying two algebraic fractions you should first:

- multiply out both numerators
- find a common denominator
- factorise everything and cancel
- add the denominators

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{2}{x} \cdot \frac{x}{3}$

ANSWER $\frac{2}{3}$

2. $\frac{a}{b} \cdot \frac{b}{a}$

ANSWER 1

3. $\frac{3}{x} \cdot \frac{x^2}{6}$

ANSWER $\frac{x}{2}$

4. $\frac{x}{2} : \frac{x}{4}$

ANSWER 2

5. $\frac{a}{3} : \frac{a}{6}$

ANSWER 2

6. $\frac{2x}{y} \cdot \frac{y}{4}$

ANSWER $\frac{x}{2}$

7. $\frac{5}{a} : \frac{10}{a^2}$

ANSWER $\frac{a}{2}$

Medium · the standard question

7 PROBLEMS

1. $\dfrac{x + 1}{x} \cdot \dfrac{x}{x + 1}$

ANSWER 1

2. $\dfrac{x^2 - 1}{x} \cdot \dfrac{x}{x - 1}$

ANSWER $x + 1$

3. $\dfrac{a^2}{b} : \dfrac{a}{b^2}$

ANSWER ab

4. $\dfrac{x - 2}{3} \cdot \dfrac{6}{x - 2}$

ANSWER 2

5. $\dfrac{x^2 - 9}{x} \cdot \dfrac{x^2}{x + 3}$

ANSWER $x(x - 3)$

6. $\dfrac{2a}{a + b} : \dfrac{4a^2}{a^2 - b^2}$

ANSWER $\dfrac{a - b}{2a}$

7. $\dfrac{x^2 + 2x}{x - 3} : \dfrac{x + 2}{x^2 - 9}$

ANSWER $x(x + 3)$

Hard · stretch and reason

7 PROBLEMS

1. $$\frac{x^2 - 4}{x^2 + 4x + 4} \cdot \frac{x + 2}{x - 2}$$

ANSWER $1, x \neq \pm 2$

2. $$\frac{a^3 - b^3}{a^2 - b^2} : \frac{a^2 + ab + b^2}{a + b}$$

ANSWER $1, a \neq \pm b$

3. $$\frac{x^2 - 5x + 6}{x^2 - 4} \cdot \frac{x^2 + 2x}{x - 3}$$

ANSWER x

4. $$\frac{2x^2 + x - 1}{x^2 - 1} \cdot \frac{x + 1}{2x - 1}$$

ANSWER $\frac{x + 1}{x - 1}$

5. $$\left(1 - \frac{1}{x}\right) : \frac{x - 1}{x^2}$$

ANSWER x

6. $$\frac{m^2 - n^2}{m^2 + mn} : \frac{m - n}{m}$$

ANSWER $1, m \neq 0, m \neq \pm n$

7. $$\frac{x^2 - 9}{x^2 - 6x + 9} : \frac{x + 3}{x - 3}$$

ANSWER $1, x \neq 3, x \neq -3$

07 Homework

Homework — 6 problems

Algebra 8, §5, pp. 27–29. Factorise before multiplying, every time.

1. $\frac{4}{x} \cdot \frac{x^2}{8}$
2. $\frac{a}{5} : \frac{a^2}{10}$
3. $\frac{x^2 - 25}{x} \cdot \frac{x}{x - 5}$
4. $\frac{x + 4}{x^2 - 16} \cdot (x - 4)$
5. $\frac{3a}{a - b} : \frac{9a^2}{a^2 - b^2}$
6. $(\frac{2x}{3y})^3$